Wavelet Denoising

VisuShrink: Soft Thresholding with the Universal Threshold

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 5

Based on D. L. Donoho and I. M. Johnstone, "Ideal spatial adaptation by wavelet shrinkage", *Biometrika* 81 (1994), 425–455

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1 Why Wavelets

A moving average smooths every part of a series in the same way. That works when the underlying signal changes slowly, but financial prices also jump: a news release or a large order moves the price in seconds. Uniform smoothing rounds such a jump off over the whole averaging window. **Wavelet denoising** adapts to the signal instead: it smooths heavily where the series is calm and keeps sharp features where they occur. It does so by looking at the series one time scale at a time.

2 The Wavelet Transform in Brief

Fourier analysis describes a series as a sum of sines and cosines. Each of these has an exact frequency but lasts for the whole sample, so a Fourier coefficient says *how much* of a frequency is present, not *when*. A jump at 10:15 spreads over every Fourier coefficient.

A **wavelet** is a short, oscillating wave that dies out quickly: it is localised in time as well as in frequency. The wavelet transform compares the series with copies of one **mother wavelet** ψ that are shifted along the time axis and stretched by powers of two,

$$\psi_{j,k}(t) = 2^{-j/2}\psi\big(2^{-j}t - k\big),$$

where k sets the position and j the scale. Each coefficient $d_{j,k}$ measures how much the series oscillates at scale 2^j around time $2^j k$.

In frequency terms, each level j acts as a **band-pass filter** covering one octave: the finest level picks up the highest frequencies, and each coarser level the octave below. Whatever is left below the coarsest level forms the **approximation**, a low-pass version of the series. Short wavelets at fine scales give precise timing but only a rough frequency band; long wavelets at coarse scales give a narrow frequency band but less precise timing. This is the compromise wavelets make that Fourier analysis does not. For 1-minute bars the levels cover these bands:

Coefficients	Periods covered	Captures
d_1	2–4 minutes	Bar-to-bar noise, bid–ask bounce
d_2	4–8 minutes	Very short-term fluctuations
d_3, d_4	8–32 minutes	Short-term moves
$d_5, \ldots$	32 minutes and up	Intraday trends
Approximation	Longest	The overall level of the day

Common choices of mother wavelet are the Daubechies family (`db4`, `sym8`); longer wavelets are smoother and separate the frequency bands more cleanly. Two facts matter for denoising. The transform is **exactly invertible**, so nothing is lost; and it **preserves the sum of squares**, so independent Gaussian noise of variance σ^2 in the

data becomes independent Gaussian noise of the same variance in every coefficient.

3 Denoising by Thresholding

Write the observed series as signal plus noise, $y_t = f_t + \sigma\varepsilon_t$ with ε_t independent $N(0,1)$. Typical signals are **sparse** in the wavelet domain: a smooth trend, or a trend with a few jumps, is described by a handful of large coefficients, while the noise spreads evenly over all of them. A jump, for example, only affects the few coefficients whose wavelets overlap it at each scale. So we keep the large coefficients and remove the small ones, using one of two rules with a threshold λ:

$$\text{hard: } \eta_\lambda(d) = d \cdot \mathbf{1}\{|d| > \lambda\}, \qquad \text{soft: } \eta_\lambda(d) = \operatorname{sign}(d)\max(|d| - \lambda, 0).$$

Hard thresholding keeps the surviving coefficients unchanged; soft thresholding also shrinks them towards zero by λ, which gives smoother output but biases large coefficients downwards.

4 VisuShrink

Donoho and Johnstone's **VisuShrink** fixes both the rule and the threshold, so there is nothing to tune:

Transform
Compute the wavelet coefficients of $y_1, \ldots, y_n$.

Estimate the noise level
The finest details d_1 are almost pure noise, so $\hat\sigma = \text{median}(|d_1|)/0.6745$. The constant is the median of $|Z|$ for a standard normal Z; the median makes the estimate robust to the few large coefficients that are signal.

Soft-threshold
Apply soft thresholding to all detail coefficients with the **universal threshold** $\lambda = \hat\sigma\sqrt{2\ln n}$. The approximation is left unchanged.

Invert
Transform back to obtain the denoised series.

Why $\sqrt{2\ln n}$. The largest of n independent $N(0,\sigma^2)$ values is close to $\sigma\sqrt{2\ln n}$. A threshold at that level therefore removes essentially every coefficient that is pure noise, so the output is visually noise-free — hence the name. The price is that small genuine features below the threshold are removed too.

5 Applying VisuShrink to Minute Bars

With PyWavelets the whole method is a few lines:

```
import numpy as np, pywt

coeffs = pywt.wavedec(prices, "db4")                        # transform
sigma = np.median(np.abs(coeffs[-1])) / 0.6745              # noise level from finest details
lam = sigma * np.sqrt(2 * np.log(len(prices)))              # universal threshold
coeffs = [coeffs[0]] + [pywt.threshold(c, lam, mode="soft") for c in coeffs[1:]]
denoised = pywt.waverec(coeffs, "db4")[:len(prices)]        # invert
```

In practice three points matter:

- **Jumps stay sharp.** A jump produces large coefficients that survive the threshold, so it is kept in the denoised series where a moving average would round it off.
- **Soft thresholding is biased.** Shrinking every surviving coefficient by λ pulls neighbouring flat levels towards each other. Hard thresholding at the same λ avoids this bias and is often more accurate; soft gives a smoother result.
- **The transform looks ahead.** Denoising a whole day uses future prices at every point, so it describes the day but cannot be traded. A causal version denoises a rolling window of past bars and keeps only the last value; it is usable in real time but less accurate, because wavelet estimates are weakest at the edge of the window.